SimLKAS: A Simulation-Based Framework for Assessing the Performance of Lane Keeping Assistant Systems under Diverse and Challenging Environments

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Introduction

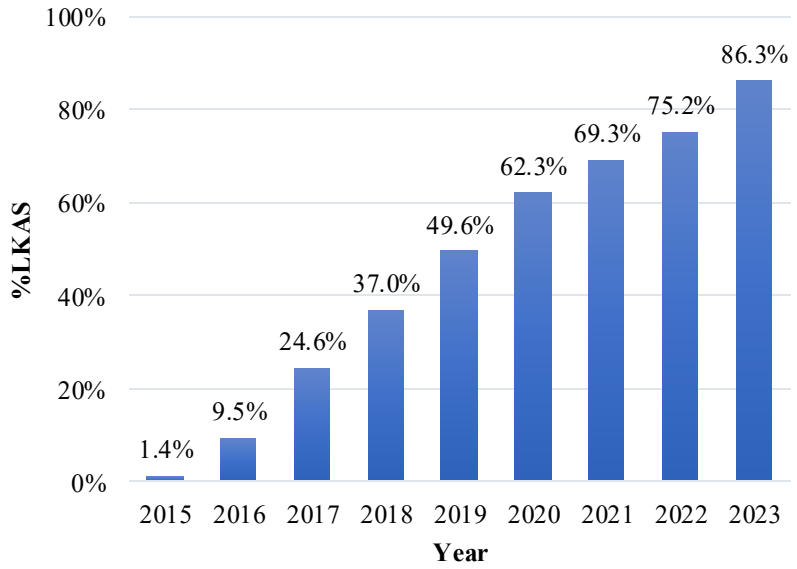
Lane Keeping Assistant Systems (LKAS), also known as **lane centering assist**, use vision-based sensors and advanced image processing algorithms to detect lane markings and provide automated steering control, helping vehicles stay centered within their lane.

It enhances driving convenience and safety by reducing driver workload and minimizing lane departure risks.

The market penetration of LKAS is advancing rapidly and steadily.



Source: https://www.honda-mideast.com/en-sa/safety/Lane-Keep-Assist-System



Source: MITRE. (2024). PARTS: Market Penetration of Advanced Driver Assistance Systems (ADAS).

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Research Motivation

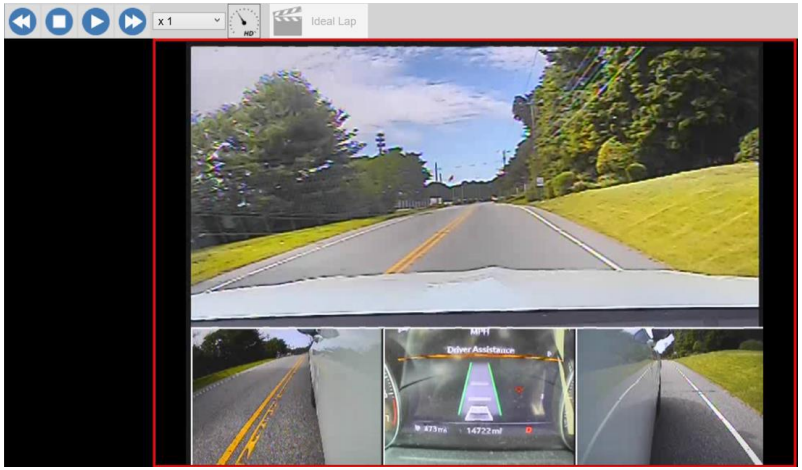
Industry standards focus on controlled test scenarios with predefined criteria.

- ISO 11270: Tests on straight/curved roads at 20–22 m/s; lateral deviation limits: 0.4 m (light vehicles), 1.1 m (heavy vehicles).

Current research rely on real-world testing, which is time-consuming, costly, less controllable, and potentially hazardous.

- Evaluating the impact of pavement marking width, color, contrast, continuity, and retroreflectivity on LKAS performance.
- Evaluation metrics : whether the LKAS can detect the left, right, or both lane markings.

Simulation-based assessment provides a controlled, scalable, reproducible, and structured evaluation approach.



Source: Shaon, M. R., Orlova, E., & Jackson, E. (2024). Automated Vehicle and Pavement Marking Evaluation in Connecticut.

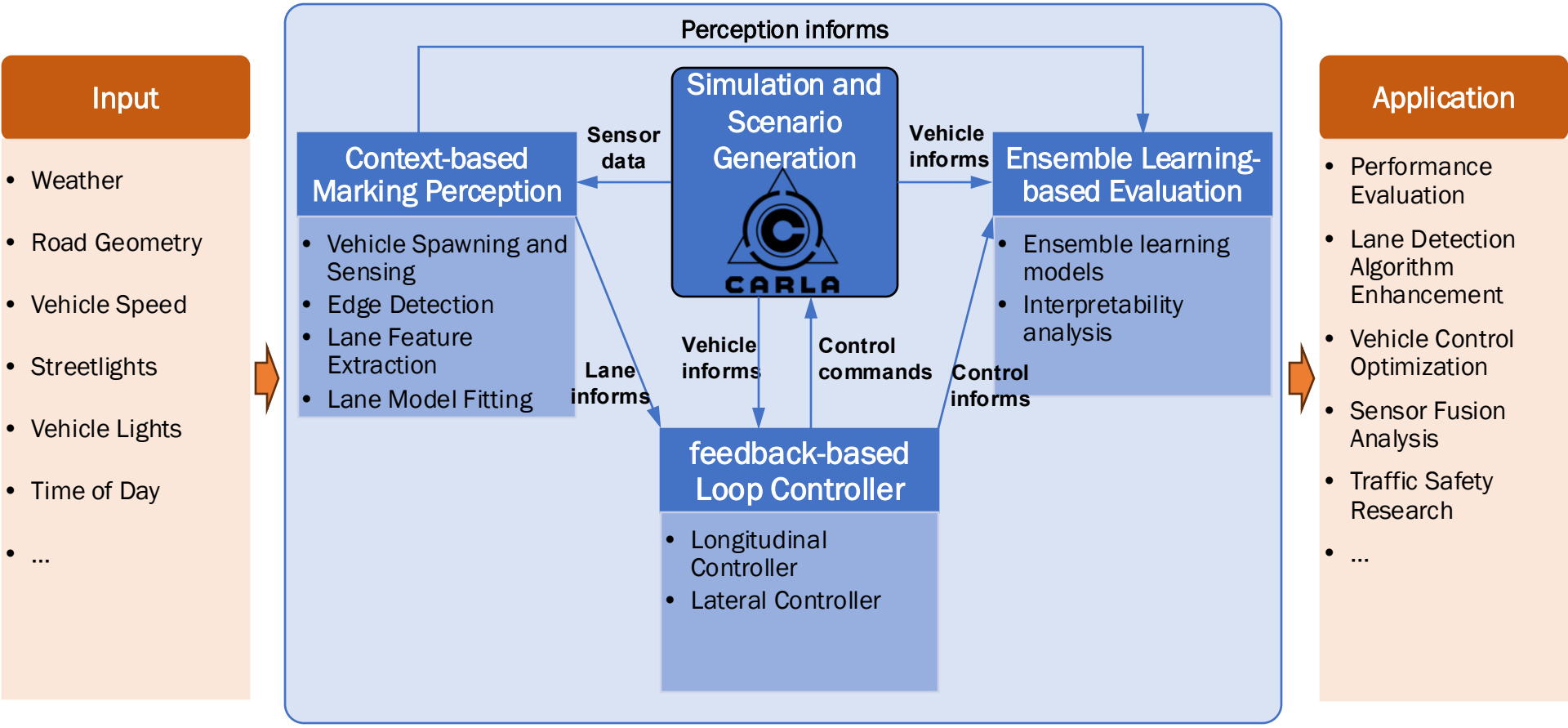


Source: Neumeister, D. M., & Pape, D. B. (2019). Automated vehicles and adverse weather.

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Framework Design

CARLA: Open-source simulation platform. Provides high-fidelity environments for testing AV technologies.



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Pavement Marking Oriented Perception Module

Preprocessing and Edge Detection

- Grayscale Conversion

$$I(x,y) = 0.299R(x,y) + 0.587G(x,y) + 0.114B(x,y)$$

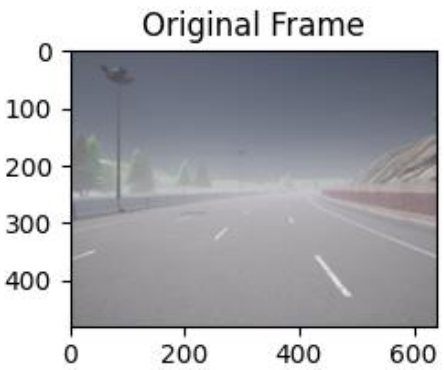
- Canny Edge Detection

Identify the edges based on the gradients of image intensity.

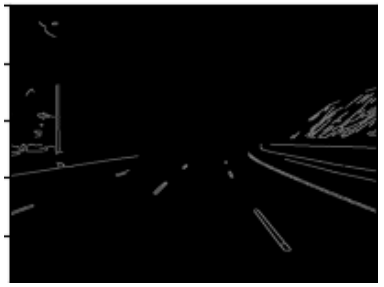
- Bird's Eye View Transformation

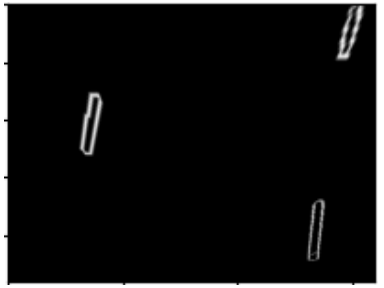
Transform the front-facing view to bird's eye view for the following process.

Define the region of interest and exclude the top of the image to remove irrelevant parts.







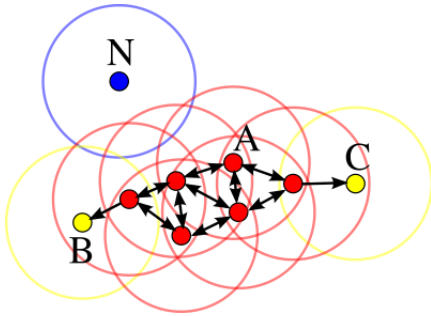


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Pavement Marking Oriented Perception Module

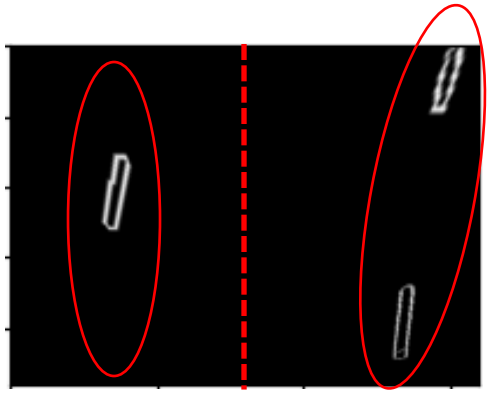
Lane Feature Extraction

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise)



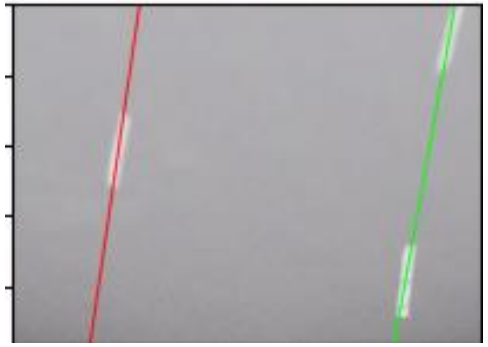
A clustering algorithm that groups dense regions of data while ignoring noise.

- ➢ Finds core points with enough neighbors within a distance ε.
- ➢ Expands clusters by connecting nearby points.
- ➢ Identifies outliers (noise points) that don't fit into clusters.



Lane Model Fitting

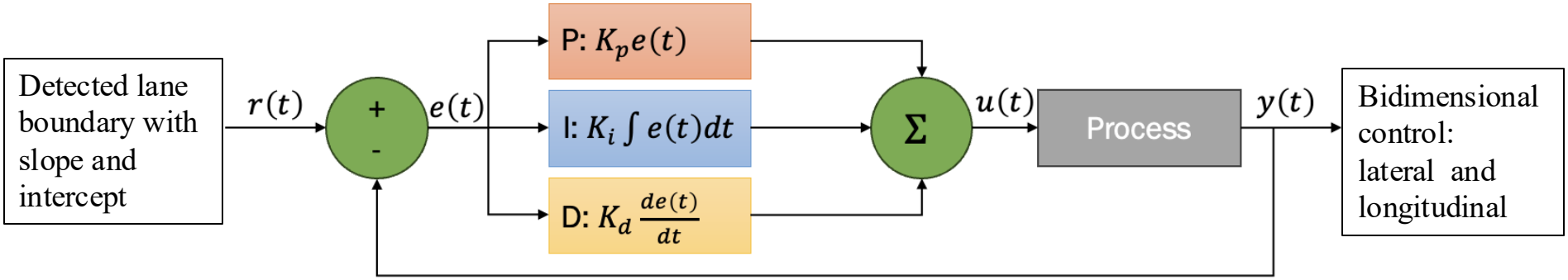
- linear regression to obtain the lane marking coefficients (slope and interception)



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Vehicle Control Module

Proportional-Integral-Derivative (PID) Control



$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

- $K_p e(t)$: Reacts to present error. A higher K_p results in a stronger response but may lead to oscillations.
- $K_i \int e(t)dt$: Reacts to past error and fixes steady-state drift. A higher K_i helps reduce long-term drift but can cause overshoot.
- $K_d \frac{de(t)}{dt}$: Reacts to future error and help smooths changes. A higher K_d helps prevent overshooting but can amplify noise.

Vehicle Control Module

Longitudinal PID Control

Input: current speed. Output: throttle/brake commands.

$$e_{speed} = v_{target} - v_{current}$$

Lateral PID Control

Input: detected lanes from the perception module. Output: steering commands.

$$e_{combined} = \gamma e_{lateral}^{meter} + \delta e_{angle}$$

Lateral error:

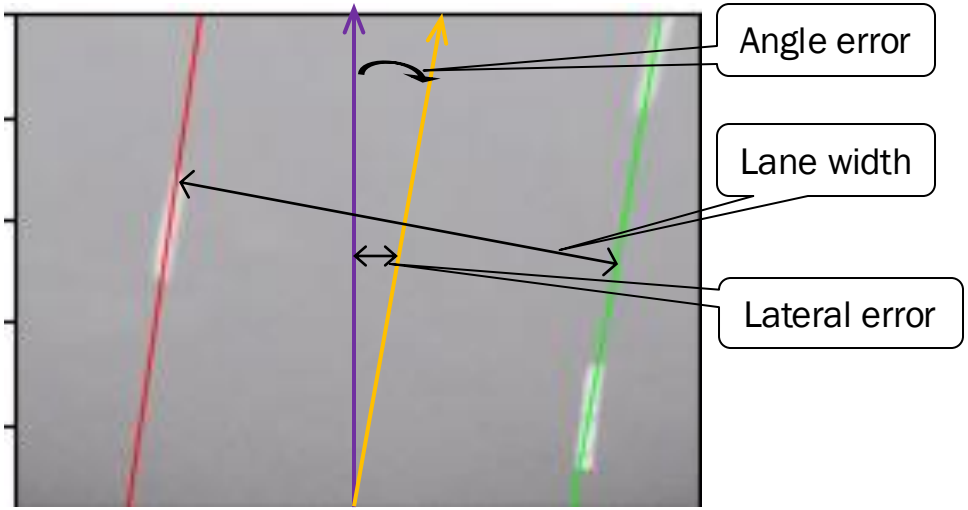
$$e_{lateral} = x_{center} - x_{vehicle}$$

Lateral error in meters:

$$e_{lateral}^{meter} = e_{lateral} \times \left(\frac{W_{lane}^{meter}}{W_{lane}^{pixel}} \right)$$

Angle error:

$$e_{angle} = \theta_{lane} - \theta_{vehicle}$$



Evaluation Module

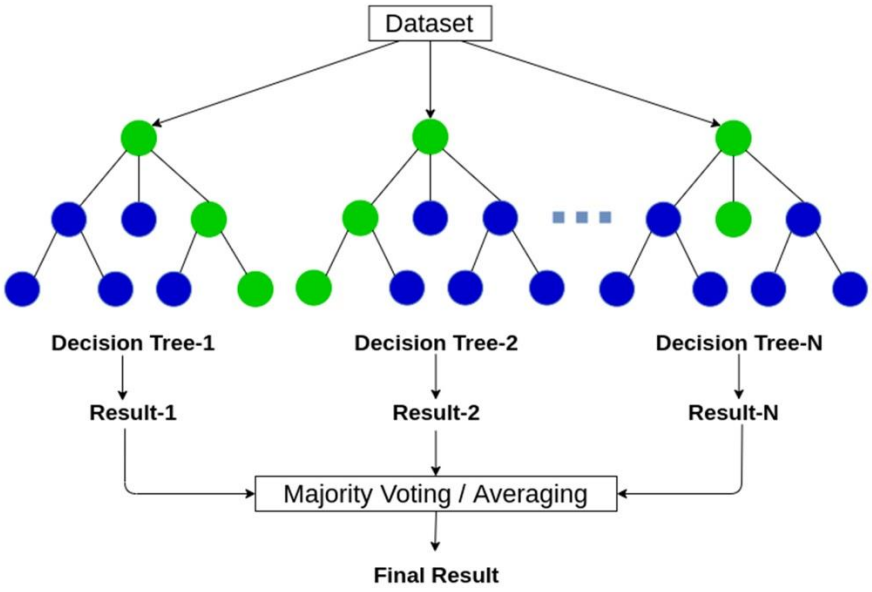
Random Forest Regression

- Ensemble Learning Method: Combines multiple decision trees to improve prediction accuracy and reduce overfitting.
- Handles Nonlinear Relationships: Captures complex patterns by averaging multiple tree predictions.
- Feature Importance Analysis: Identifies key variables that influence the target outcome.

SHAP Analysis

SHAP is a machine learning interpretability tool based on Shapley values from game theory.

- Provides Local & Global Interpretability: Helps understand both individual predictions and overall feature importance.



Source: https://anasbrital98.github.io/blog/2021/Random-Forest/.

Experimental settings

Variable	Type	Explanation
Cloudiness	Float (0, 100)	Controls the intensity of the clouds
Precipitation	Float (0, 100)	Controls the strength of the precipitation
Precipitation deposits	Float (0, 100)	Determines the creation of puddles
Wind intensity	Float (0, 100)	Controls the strength of the wind
Sun azimuth angle	Float – degrees (0, 360)	The azimuth angle of the sun
Sun altitude angle	Float – degrees (-90, 90)	Altitude angle of the sun
Fog density	Float (0, 100)	Fog concentration or thickness
Fog distance	Float – meters	Fog start distance
Wetness	Float (0, 100)	Wetness intensity
Fog falloff	Float	Density of the fog
Scattering intensity	Float	Controls how much the light will contribute to volumetric fog
Mie scattering scale	Float	Controls interaction of light with large particles
Rayleigh scattering scale	Float	Controls interaction of light with small particles
Dust storm	Float (0, 100)	Determines the strength of the dust storm weather

Experimental settings

- **Map**: Town03
- **Length**: 350m
- **Speed**: 30, 40, 50, 60 km/h
- **Weather**: Clear, middle rain, heavy rain, middle fog, heavy fog
- **Day of time**: daytime, nighttime
- **Road geometry**: straight, curved
- **Streetlights**: on, off.
- **Vehicle light**: high beam, low beam

Total of 36 scenarios. Each scenario run 3 times.





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Experimental settings

- **Map**: Town04
- **Length**: 1km
- **Speed**: 30, 50, 70, 90 km/h
- **Weather**: Clear, middle rain, heavy rain, middle fog, heavy fog
- **Day of time**: daytime, nighttime
- **Road geometry**: straight, curved
- **Streetlights**: on, off.
- **Vehicle light**: high beam, low beam

Total of 36 scenarios. Each scenario run 3 times.



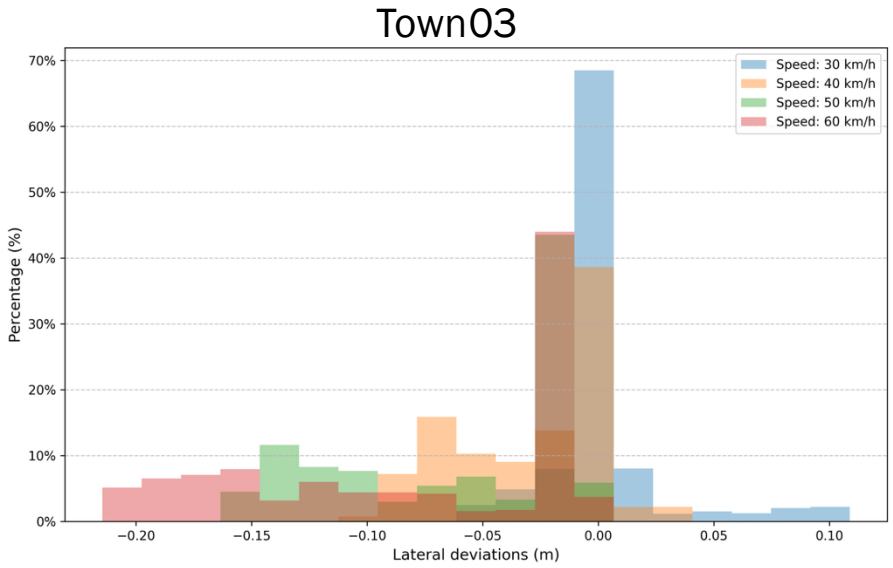


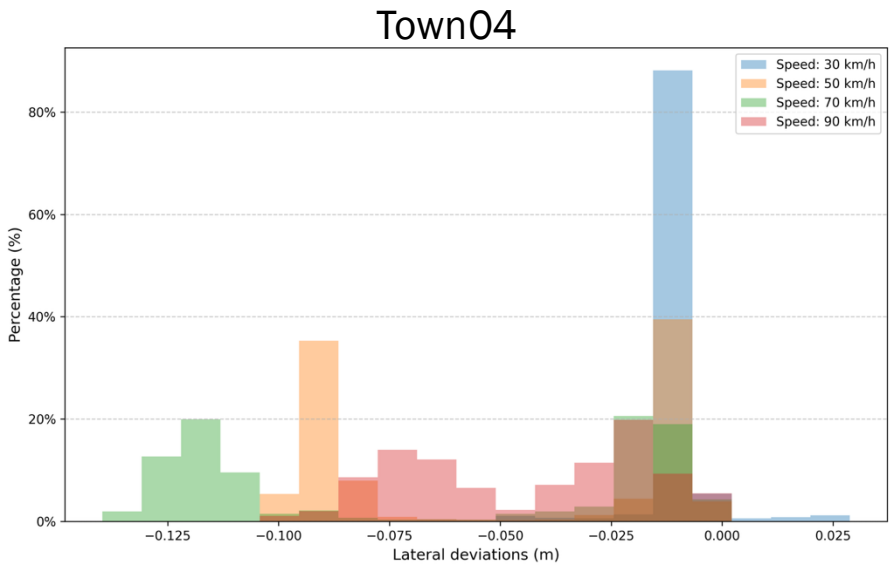
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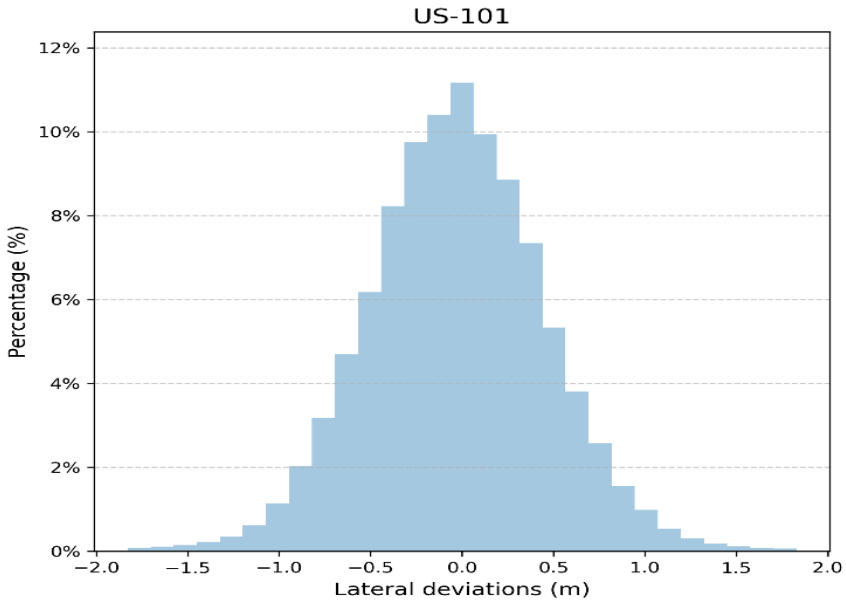
Results – Baseline Performance

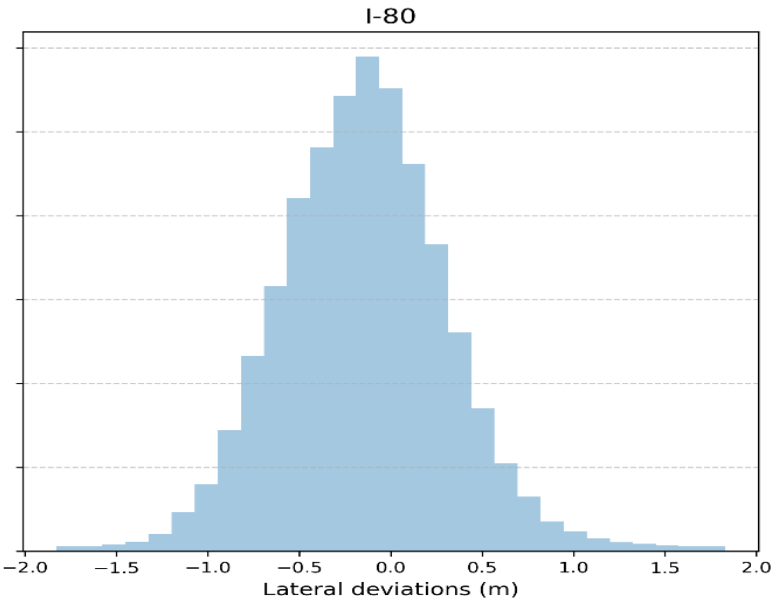


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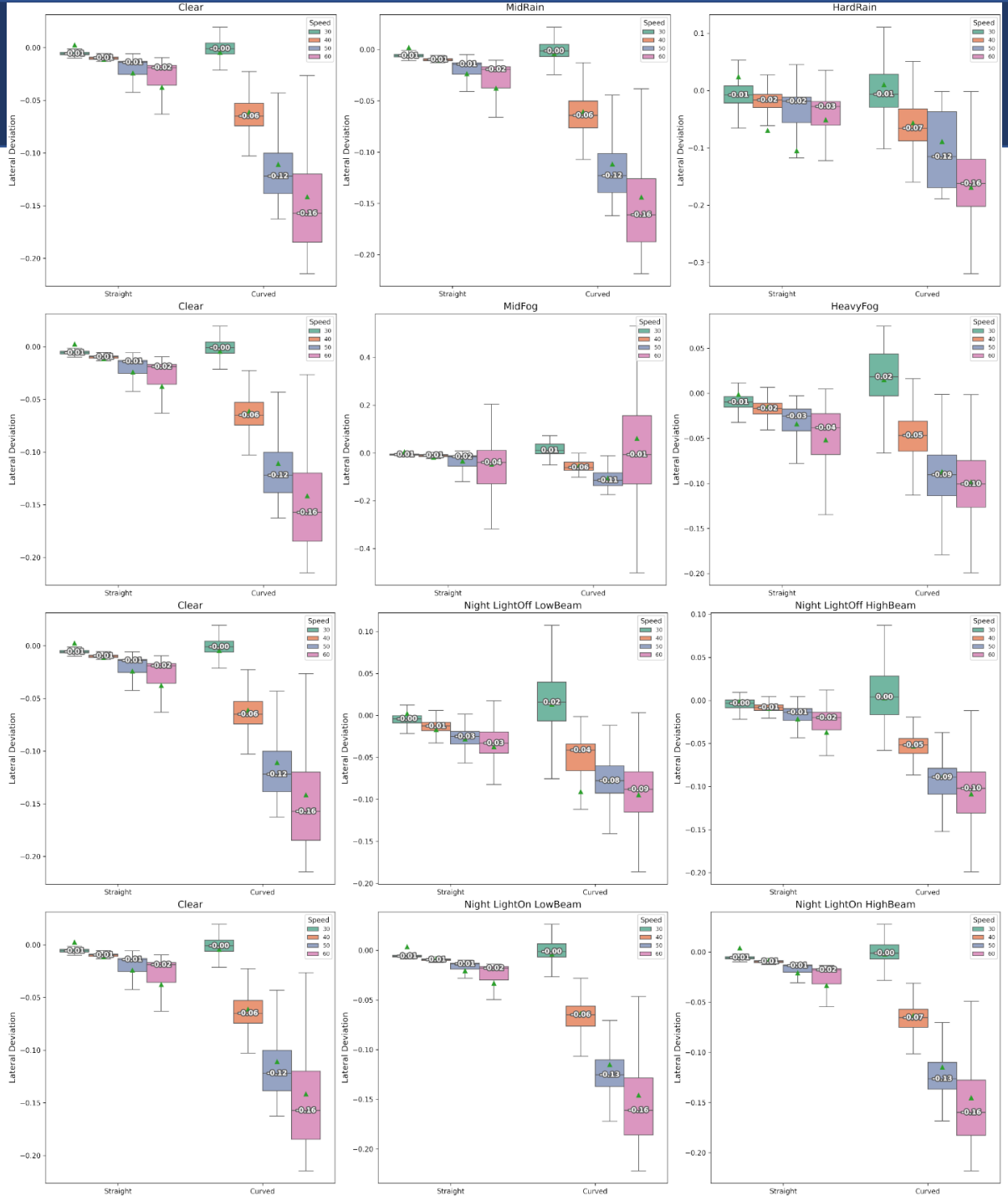






Results — Town03

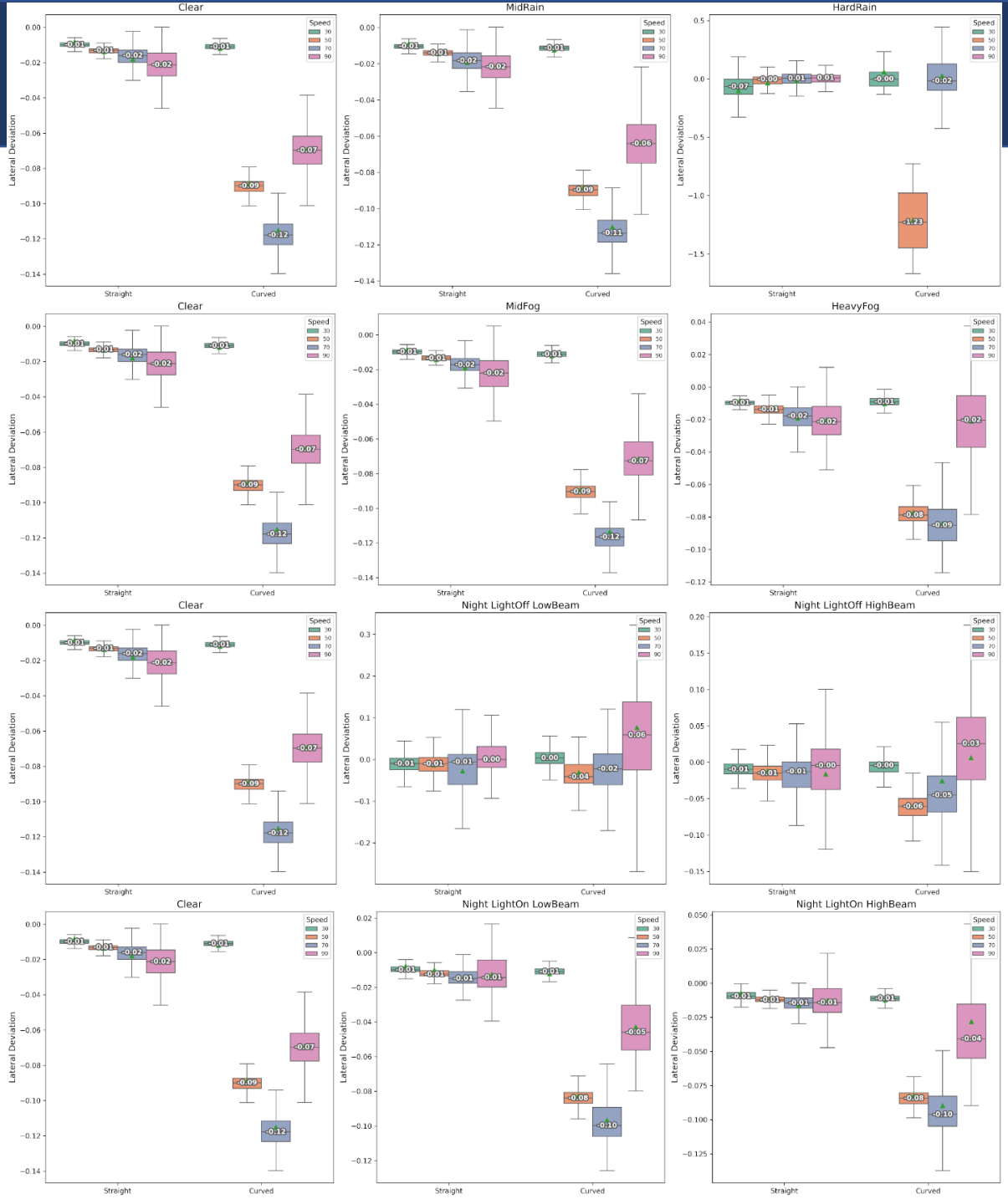
- Adverse weather conditions negatively impact LKAS performance, particularly at higher speeds and on curved roads.
- Heavy rain degrades LKAS performance more than heavy fog.
- Performance more stable in low speeds.
- Adequate lighting, including high beam and streetlights on, significantly enhances LKAS performance at night.
- LKAS remains effective in moderate weather conditions but requires further optimization for extreme conditions.



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Results — Town04

- Adverse weather conditions negatively impact LKAS performance, particularly at higher speeds and on curved roads.
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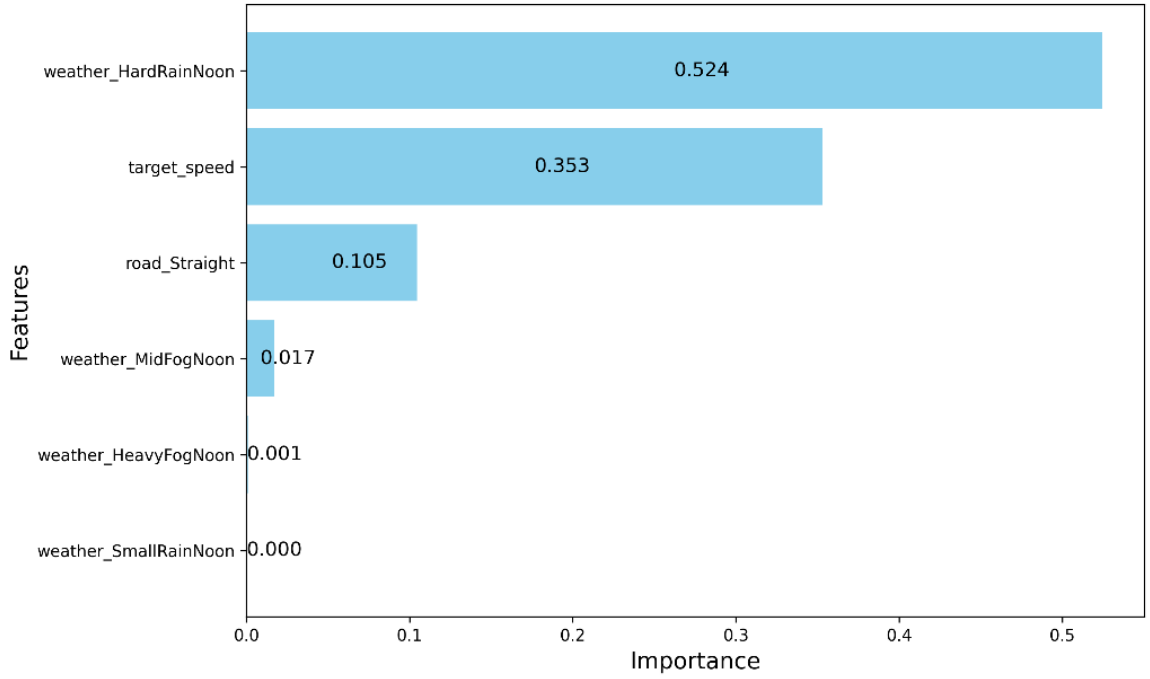


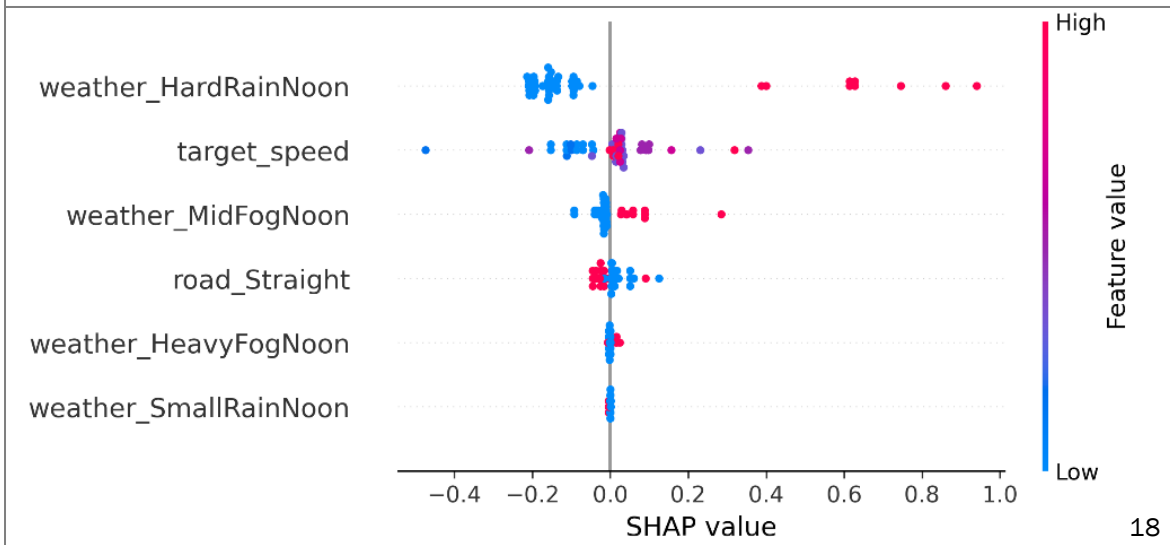
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Results —— RF + SHAP

Daytime Model
R^2: 0.63
MSE: 0.090

- Heavy rain (red dots) consistently shifts SHAP values in a positive direction, demonstrating its significant negative impact on LKAS performance.
- Higher speed increase SHAP values, meaning it positive influence max lateral deviation.



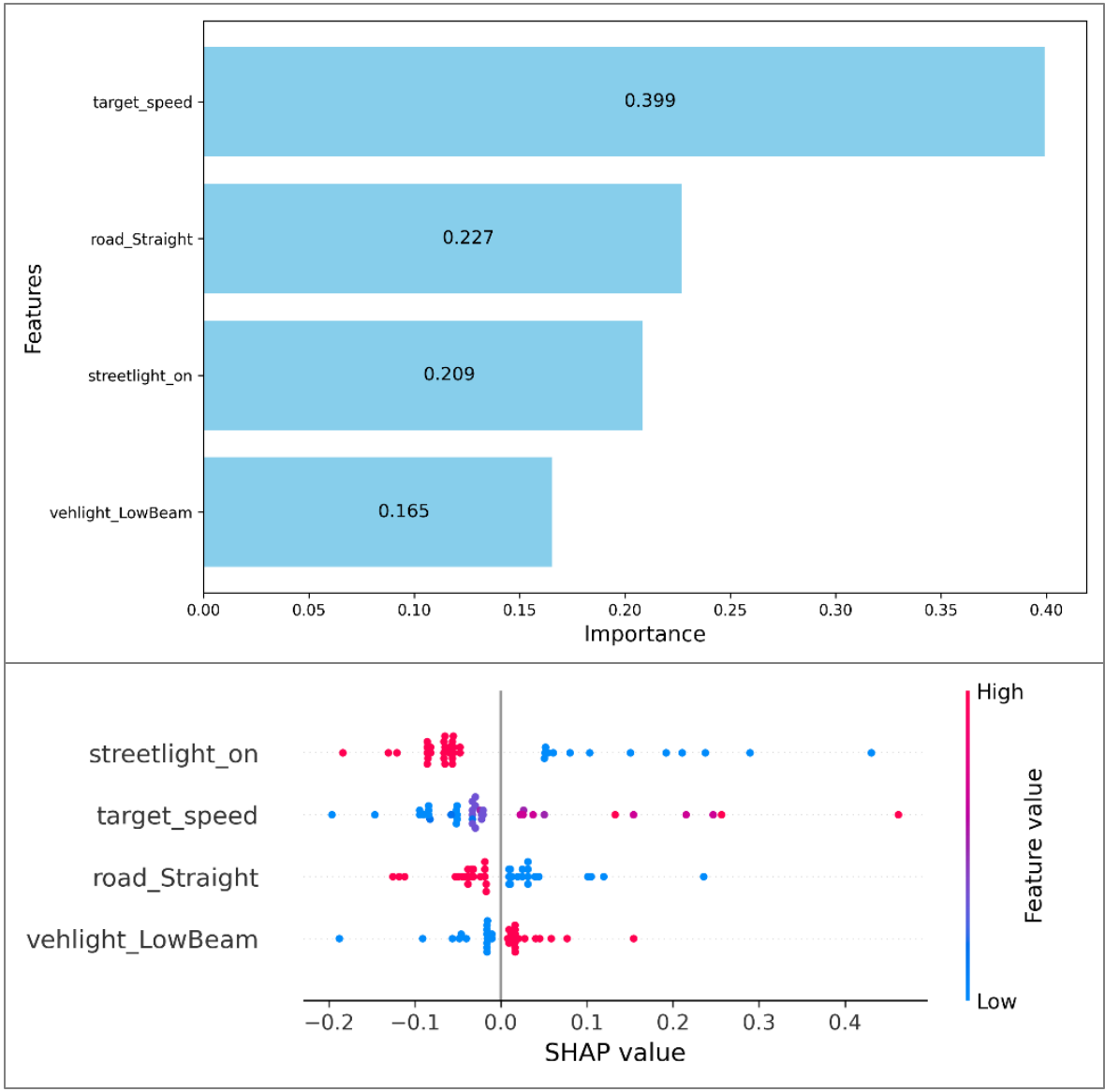


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Results — RF + SHAP

Nighttime Model
R^2: 0.31,
MSE: 0.080

- Streetlight on (red dots) reduce max lateral deviations.
- Higher speed strongly correlate with higher max lateral deviations.
- Road geometry has a greater influence on nighttime performance.
- Using high beams corresponds to improved lane-keeping performance.



Conclusions

The proposed framework ensures controllable, standardized, repeatable, and cost-efficient performance comparisons. It also provides a scalable modular structure for evaluating critical LKAS techniques under diverse scenarios.

- During daytime, heavy rain and high speeds are the dominant factors to increased lateral deviation.
- At night, visibility-related conditions such as streetlights and headlight settings play a crucial role.
- Curved roads consistently present additional challenges across all scenarios, further increasing lane deviation compared to straight roads.

Future work

- Develop deep learning-based lane detection algorithms.
- Develop reinforcement learning-based control strategies.
- Develop a co-simulation platform with SUMO for more comprehensive performance evaluations, including safety analysis in complex traffic scenarios.
- Conducting field tests is still essential to assessing LKAS performance.

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Thank you & Questions?

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